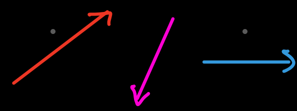
By Clifford (1878) Developing Geometric Algebra from Highschool Algebra

there are many types of flavors in GA

Vanilla Geometric Algebra

Vectors



is the simplest Geometric object that you can do Algebra with

every vector starts in the origin

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \qquad x_\mu \quad \vec{x} = x_1 \hat{x} + x_2 \hat{y} + x_3 \hat{z} = (x_1, x_2, x_3)$$

What is space?

What regions we consider to be SPACE?

$\downarrow$

What sets of vectors do we consider to be a space

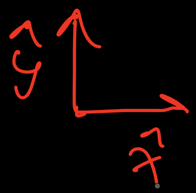
The set of linear combinations of n linear independent basis vectors

$\{u_1, u_2\}$ $\rightsquigarrow$ any sum of any vectors in the space must be still in the space

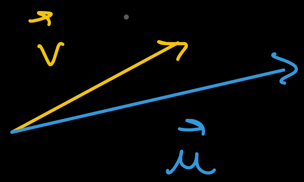
& the same for scaling

Describing a space with few vectors

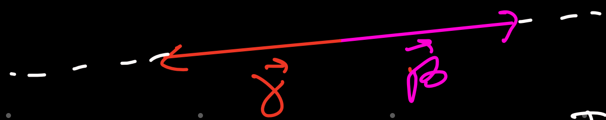
all the vector that are of the form $a\hat{x} + b\hat{y}$



are all the vectors of the plane



$a\vec{v} + b\vec{u}$ → also allows any vector on the plane



$a\vec{\gamma} + b\vec{\beta}$ → Does NOT allow to reach any vector in the plane

But it does in the line

$a\hat{x} + b\hat{y}$ → Linear Combinations

The vectors that can be written as a linear combination of the vectors in the set $\{\hat{x}, \hat{y}\}$ are called the Span

$\eta = \{\hat{x}, \hat{y}\}$ $\quad \text{span}(\eta) =$ <u>xy plane</u>

Is the span of a set of vectors always a space?

$S = \{\vec{v}_1, \vec{v}_2, \vec{v}_3, \dots, \vec{v}_n\}$

a vector in the span(S)

$$a_1\vec{v}_1 + a_2\vec{v}_2 + a_3\vec{v}_3 + \dots a_n\vec{v}_n$$
$$+\ b_1\vec{v}_1 + b_2\vec{v}_2 + b_3\vec{v}_3 + \dots b_n\vec{v}_n$$

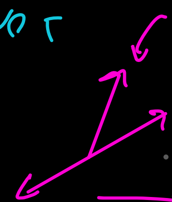
$$(a_1+b_1)\vec{v}_1 + (a_2+b_2)\vec{v}_2 + (a_3+b_3)\vec{v}_3 + \dots (a_n+b_n)\vec{v}_n$$

↪ its still in the span(S)

This way the SPAN is always a SPACE

If removing vector from a set changes its SPAN the set is linearly Independent

If removing any vector does not it is linearly Dependent



→ Linearly Dependent

A set of linearly Independent Vectors are a Base for a SPACE

that span a SPACE

A VECTOR is an element of linear SPACE

functions are linear spaces

lines too
so they are vectors

Lines Being linear Spaces

$$ax + by + c = 0 \longleftrightarrow a e_1 + b e_2 + c e_0$$

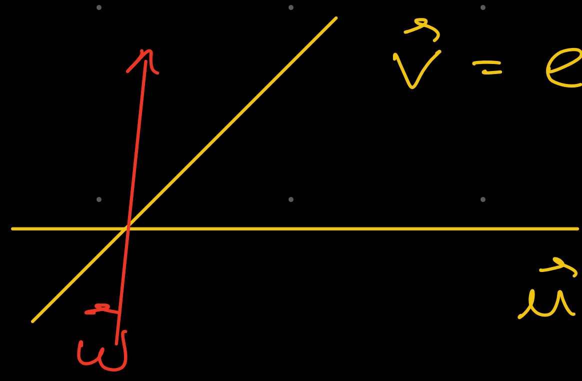
as e_1, e_2, e_3 are basis for the line vector space
each of them must represent a line

$e_1 \Rightarrow x = 0$

$e_2 \Rightarrow y = 0$

$e_0 \Rightarrow 1 = 0$ WTF $\rightsquigarrow$ like a line to infinity

e_0 is the line at Infinity not $1=0$



$\vec{v} = e_1 - e_2 + e_0$

$\vec{u} = e_2 + e_0$

$\vec{u} + \vec{v} = \vec{w} \Rightarrow e_1 - \cancel{e_2} + e_0 + e_0 + \cancel{e_2}$

$$e_1 + 2e_0$$

When adding two lines the sum passes through their intersection